

ABHIMUNNYA DEY

Software Engineer | Full Stack Developer

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SUMMARY

Software Engineer with hands-on experience in full-stack development using **Java**, **JavaScript**, **Python**, and modern frameworks including **Spring Boot** and **React**. Skilled in **Data Structures & Algorithms**, **Object-Oriented Programming**, and **REST API design**, with a track record of building **scalable backend systems** and **responsive web interfaces**. Currently gaining professional experience as a Software Engineer Intern, contributing to **full-stack development**.

SKILLS

Languages: Java · JavaScript · Python

Frameworks: Spring Boot · Spring Data JPA · Spring Security · JWT · React

Databases: PostgreSQL

Concepts: Data Structures & Algorithms · OOP · Collections Framework · Full-Stack Development · REST API Design

Tools: Git · Postman

EXPERIENCE

Omnivera Technologies Pvt. Ltd

Software Engineer Intern

Remote

07/2026 – Present

- Developed responsive and tablet-optimized web interfaces using React.js, JavaScript, and Tailwind CSS, applying reusable component-based architecture, responsive design, and maintainable coding practices.
- Currently working on backend and database development using Node.js, Express.js, and MongoDB, with exposure to RESTful API design, authentication/authorization, data modeling, and frontend-backend integration.

PROJECTS

Offline UPI Payment Mesh – Distributed Encrypted Transaction Relay System

Github

Distributed backend system that settles encrypted offline UPI payments through a Bluetooth mesh relay network with exactly-once transaction guarantees.

- Backend system that enables UPI payments without internet by relaying encrypted payment packets across a Bluetooth mesh network until a bridge node settles the transaction against a PostgreSQL ledger.
- Hybrid RSA-OAEP + AES-GCM encryption, atomic idempotency via ConcurrentHashMap putIfAbsent, optimistic locking via @Version, gossip protocol with TTL-based propagation, and a 5-stage ingestion pipeline covering deduplication, decryption, freshness validation, and atomic settlement.
- Eliminates 100% of duplicate settlements, replay attacks, and balance corruption under concurrent multi-node delivery – solving complete payment failure in zero-connectivity environments.

AskMyResume – AI-Powered Interactive Portfolio Chatbot

Github

Full-stack AI chatbot that answers questions about my background, using structured resume data and strict LLM grounding to prevent hallucinated answers.

- Full-stack AI-powered portfolio chatbot that lets visitors interactively ask about my skills, experience, and projects instead of reading a static resume, built with a FastAPI REST backend and a React SPA frontend.
- Implemented LLM-based resume parsing using the Groq API, validating extracted data against a strict Pydantic schema; added startup-time in-memory caching to avoid redundant LLM calls, CORS-configured API access, and structured error handling for malformed AI responses and upstream API failures.
- Solves the problem of static, low-engagement resumes by offering a conversational, hallucination-resistant Q&A experience through a responsive React + Tailwind CSS interface with real-time Markdown-formatted answers.

EDUCATION

Bengal Institute Of Technology

2023 – 2027

B.Tech in Computer Science · CGPA: 7.15

LANGUAGES

Bengali Native

English Proficient

Hindi Proficient

German Beginner